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Sampling

When generating text the language model gives several different options for which word could come next in the generated text — which one to choose?

You will need

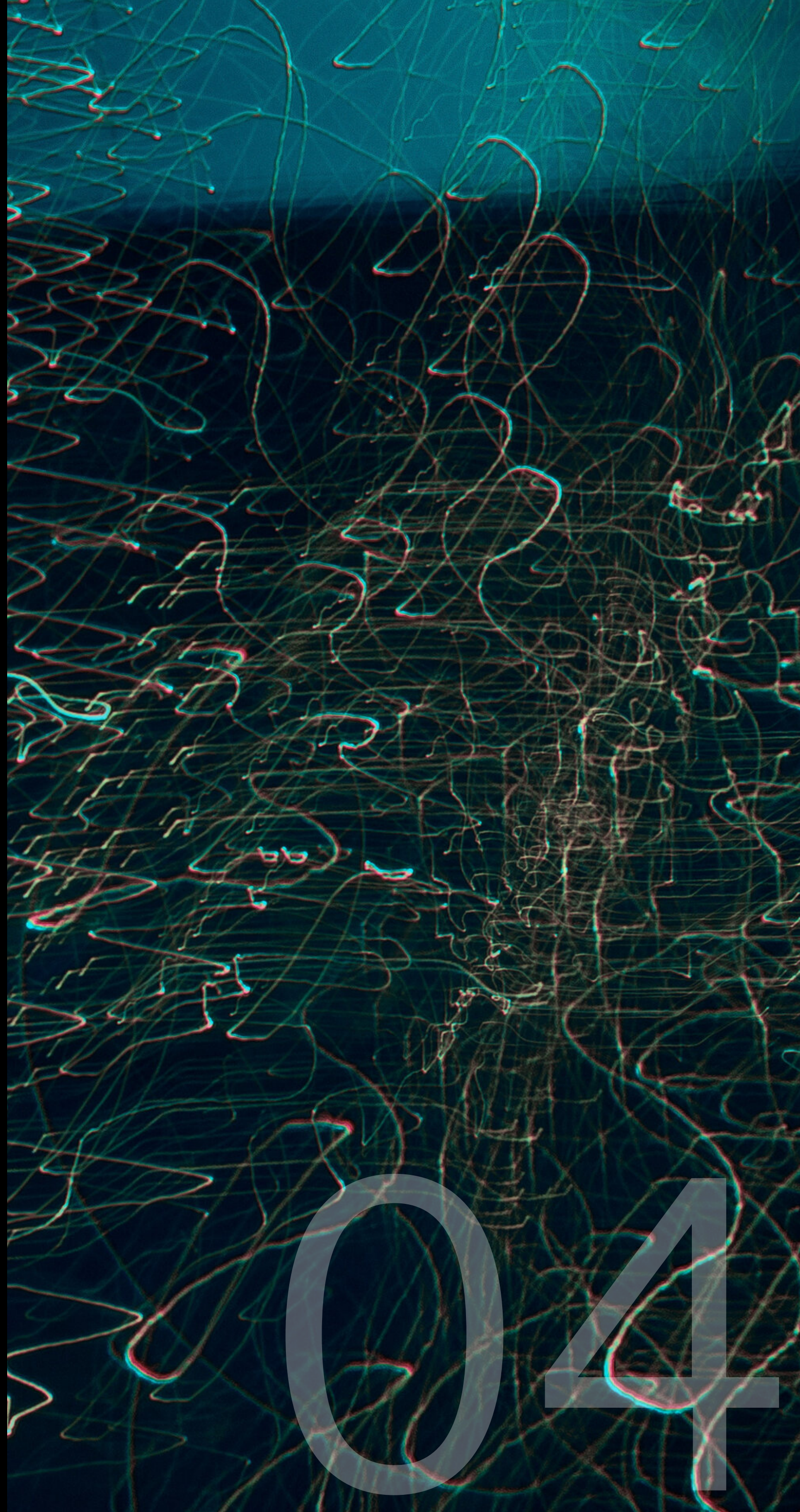
- a completed model from an earlier lesson
- pen, paper & dice as per *Basic Generation*

Your goal

To generate text (with the same model) using at least two different temperature values and at least two different truncation strategies. **Stretch goal**: design and evaluate your own truncation strategy.

Key idea

There are lots of different sampling algorithms — ways to select the next word during text generation. Each strategy has different strengths and weaknesses, and can significantly influence the generated text even if the rest of the model is identical.



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Temperature control

The temperature parameter (a number) controls the randomness by adjusting the relative likelihood of probable vs improbable words. The higher the temperature, the more uniform the distribution becomes, increasing randomness and allowing more sampling from unlikely words.

Algorithm

1. when sampling the next word, divide all counts by temperature value (round down, min 1)

Example

If the counts in a given row are:

	`spot`	`run`	`jump`	`.`
`see`	4	2	1	1

1. if **temperature = 1**: use counts as-is (4, 2, 1, 1)
   - `spot` is 2x as likely as `run`, 4x as likely as `jump` or `.`
2. if **temperature = 2**: divide counts by 2 ⟶ (2, 1, 1, 1)
   - `spot` still most likely, but only 2x as likely as others
3. if **temperature = 4**: divide counts by 4 ⟶ (1, 1, 1, 1)
   - all words equally likely

Truncation strategies

Truncation narrows the viable “next word options” by ruling out some options. Any truncation strategy can be combined with temperature control.

Greedy sampling

1. find current word’s row
2. select the word with the highest count
3. if there’s a tie, roll dice to choose equally among the most likely options

Haiku sampling

1. track syllables in current line (5-7-5 pattern)
2. roll dice to select next word as normal
3. if selected word exceeds line’s syllable limit, re-roll
4. start new line when syllable count reached

Non-sequitur sampling

1. find current word’s row
2. pick the column with the lowest (non-zero) count
3. if there’s a tie, roll dice to choose equally among the least likely options

No-repeat sampling

1. track all words used in current sentence
2. roll dice to select next word as normal
3. if word already used, re-roll
4. if no valid options remain, insert `.` and continue

Alliteration sampling

1. note first letter/sound of previous word
2. if any next-word options start with same letter/sound, sample only from those alliterative options
3. otherwise use standard sampling